Vacuum functional and Green's functions

$$\langle x_f, t_f | x_i, t_i \rangle_J = D_J(x_f, t_f; x_i, t_i) =$$

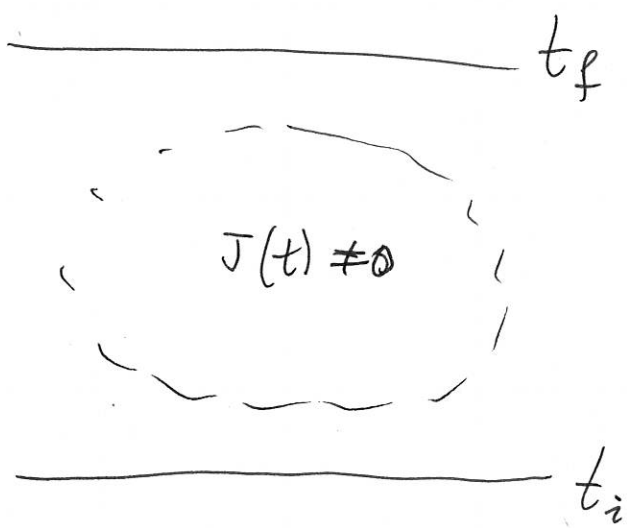
$$= N \int Dx \, e^{\frac{i}{\hbar} S_J[x]}$$

$$\downarrow$$

$$S_J = S[x] + \int_{t_i}^{t_f} dt \, J(t) x(t)$$

$$\Rightarrow \langle x_f, t_f | x_i, t_i \rangle_J = \langle x_f, t_f | \hat{T}\left\{ e^{\frac{i}{\hbar}\int dt \, \hat{x}(t) J(t)} \right\} | x_i, t_i \rangle$$

Now take the limit $t_i \to -\infty$, $t_f \to +\infty$
(while external sources $J(t) = 0$ for $|t| > \tau$)
where the limit $\tau \to \infty$ is taken at the very end



$$\lim_{\substack{t_i \to -\infty \\ t_f \to +\infty}} D_J(x_f, t_f; x_i, t_i) = \lim_{\tau \to \infty} \lim_{\substack{t_i \to -\infty \\ t_f \to +\infty}} \langle x_f, t_f | \hat{T}\left(e^{\frac{i}{\hbar}\int dt \, \hat{x} J}\right) \cdot | x_i, t_i \rangle$$

$$= \lim_{\tau \to \infty} \lim_{\substack{t_i \\ t_f}} \sum_{n,m} \underbrace{\langle x_f, t_f | n \rangle}_{\langle x_f | e^{-\frac{i}{\hbar}\hat{H} t_f} | n \rangle} \langle n | \hat{T}\left\{ e^{\frac{i}{\hbar}\int J \hat{x}} \right\} | m \rangle \underbrace{\langle m | x_i, t_i \rangle}_{\langle m | e^{\frac{i}{\hbar}\hat{H} t_i} | x_i \rangle}$$

$\Rightarrow$

$$\Rightarrow \lim_{t\to\infty} \lim_{\substack{t_i\to-\infty \\ t_f\to+\infty}} \sum_{n,m} e^{-\frac{i}{\hbar}E_n t_f + \frac{i}{\hbar}E_m t_i} \overbrace{\langle x_f|n\rangle}^{\psi_n(x_f)} \underbrace{\langle m|x_i\rangle}_{\psi_m^*(x_i)}$$

$$\cdot \langle n|\hat{T}\left\{e^{\frac{i}{\hbar}\int dt' \hat{X} J}\right\}|m\rangle$$

(Here we are using $|x,t\rangle = e^{\frac{i}{\hbar}\hat{H}t}|x\rangle = \sum_n e^{\frac{i}{\hbar}\hat{H}t}|n\rangle\underbrace{\langle n|x\rangle}_{\psi_n^*(x)} =$

$= \sum_n e^{\frac{i}{\hbar}E_n t}\, \psi_n^*(x)\, |n\rangle$

* The limit $t_i \to -\infty$, $t_f \to +\infty$ is crucial:

$$e^{-itE/\hbar} \xrightarrow[t\to\infty]{} 0 \quad \text{in the sense of a distribution}$$

and the dominant ~~the~~ non-zero contribution comes from the smallest E_n ← that is, from the ground state!

~~Thus~~ That is, the limit $(t_i \to -\infty, t_f \to +\infty)$ projects us to the ground state

$$\sum_{n,m} e^{-\frac{i}{\hbar}(E_n t_f - E_m t_i)} \ldots \longrightarrow e^{-\frac{i}{\hbar}E_0(t_f - t_i)} \ldots$$

Now can take τ-limit too

$$\langle 0|\hat{T}\left\{e^{\frac{i}{\hbar}\int_{-\infty}^{\infty} dt' J\hat{X}}\right\}|0\rangle = \lim \frac{\langle x_f, t_f | x_i, t_i\rangle_J}{\psi_0(x_f)\cdot\psi_0(x_i)}$$

LHS is independent of the end points $\Rightarrow$ RHS is independent too!

$$\langle 0|0\rangle_J = N\int Dx\, e^{\frac{i}{\hbar}S_J[x]} \qquad S_J[x] = \int_{-\infty}^{\infty} dt\,(L(x,\dot{x}) + Jx)$$

but without any end-point constraints!

If we Define $Z(J) = \langle 0|0\rangle_J = N \int Dx \, e^{\frac{i}{\hbar} S_J(x)}$

Then $$\frac{(-i\hbar)^n}{Z(J)} \frac{\delta^n Z(J)}{\delta J(t_1) \ldots \delta J(t_n)}\bigg|_{J=0} = \langle 0|\hat{T}\{\hat{X}(t_1) \ldots \hat{X}(t_n)\}|0\rangle$$

So that $Z(J)$ generates time-ordered correlation functions (Green's functions) in the vacuum.

Back to Harmonic oscillator:

$$Z(J) = N \int Dx \; e^{\frac{i}{\hbar}\int_{-\infty}^{\infty} dt \left(\frac{1}{2} m\dot{x}^2 - \frac{1}{2} m\omega^2 x^2 + Jx\right)}$$

$$\langle 0|T(\hat{X}(t_1))|0\rangle = (-i\hbar)\frac{1}{Z(J)}\frac{\delta Z(J)}{\delta J(t_1)}\bigg|_{J=0} = \frac{N\int Dx\; x(t_1) e^{\frac{i}{\hbar}S[x]}}{N\int Dx\; e^{\frac{i}{\hbar}S[x]}} = 0$$

no J here!

$$\langle T\{X(t_1)X(t_2)\}\rangle = (-i\hbar)^2 \frac{1}{Z(J)}\frac{\delta^2 Z(J)}{\delta J(t_1)\,\delta J(t_2)}\bigg|_{J=0}$$

$$\int_{-\infty}^{\infty} dt\; \frac{1}{2} m\dot{x}^2 = \frac{1}{2} m \int_{-\infty}^{\infty} dt \left[\underbrace{\frac{d}{dt}(\dot{x}x)}_{\text{can set } = 0} - x\ddot{x}\right] = \int_{-\infty}^{\infty} dt\; -\frac{1}{2} m x \frac{d^2}{dt^2} x$$

since BC does not matter anymore

$$\Rightarrow Z(J) = N\int Dx \; \exp -\frac{im}{2\hbar}\int_{-\infty}^{\infty} dt \left\{x(t)\left(\frac{d^2}{dt^2} + \omega^2 - i\varepsilon\right)x(t) - \frac{2}{m} J(t)x(t)\right\}$$

$$\left(\frac{d^2}{dt^2} + \omega^2 - i\varepsilon\right) G_F(t-t') = -\delta(t-t')$$

~~let~~ $\cancel{x(t) = \tilde{x}(t) + \frac{1}{m}\int_{-\infty}^{\infty} dt'\, G_F(t-t')\, J(t')}$

$x = x_{cl} + y$

$x_{cl}(t)$ solves $\left(m\frac{d^2}{dt^2} - m\omega^2\right)x = J$

$$\cancel{Z(J) = N\int Dy \; \exp\frac{i}{\hbar}\int dt \left(+J(t)\,x_{cl}(t)\right)}$$

$$\cancel{\cdot \exp\frac{i}{\hbar}\left\{\int dt\; y(t)\left(-\frac{m}{2}\frac{d^2}{dt} - \frac{m}{2}\omega^2\right)y(t)\right\}}$$

$$Z(J) = N \int Dx \; \exp \frac{i}{\hbar} \int_{-\infty}^{\infty} dt \; x(t) \left\{ \left(-\frac{m}{2}\right)\left(\ddot{x} + \Omega^2 x\right) + Jx \right\}$$

Eq of motion for x_{cl}, $x(t) = x_{cl}(t) + y(t)$

$$\ddot{x}_{cl} + \Omega^2 . x_{cl} = \frac{J(t)}{m}$$

$$Z(J) = N \int Dy \; \exp \frac{i}{\hbar} \int dt \; x_{cl}(t) \left\{ \left(-\frac{m}{2}\right) \frac{J(t)}{m} + J x_{cl} \right\}$$

$$\exp \frac{i}{\hbar} \int_{-\infty}^{\infty} dt \left\{ y \left(-\frac{m}{2}\right)\left(\frac{d^2}{dt^2} + \Omega^2\right) y \right\}$$

$$= Z(0) \; \exp \frac{i}{\hbar} \int dt \; \frac{1}{2} J(t) \, x_{cl}(t)$$

But $x_{cl}(t) = -\int_{-\infty}^{\infty} dt' \, G(t-t') \frac{J(t')}{m}$

(here we omit ~~the~~ homogeneous solution as boundary conditions do not matter anymore)

$$Z(J) = Z(0) \; \exp\left\{ -\frac{i}{2m\hbar} \int_{-\infty}^{\infty} dt dt' \, J(t) \, G(t-t') \, J(t') \right\}$$

Thus $\langle 0 | \hat{T} \{ x(t_1) \, x(t_2) \} | 0 \rangle = \frac{(-i\hbar)^2}{Z(0)} \left. \frac{\delta^2 Z}{\delta J(t_1) \, \delta J(t_2)} \right|_{J=0}$

$$= (-i\hbar)^2 \frac{\delta^2}{\delta J(t_1) \, \delta J(t_2)} \left. e^{-\frac{i}{2m\hbar} \int_{-\infty}^{\infty} dt dt' J(t) \, G(t,t') \, J(t')} \right|_{J=0} =$$

$$= -\hbar^2 \frac{\delta}{\delta J(t_1)} \left(-\frac{i}{2m\hbar}\right) \left[\int_{-\infty}^{\infty} dt' G(t_2, t') \, J(t') + \int_{-\infty}^{\infty} dt \; J(t) \, G(t, t_2) \right] .$$

$$\cdot \left. e^{-\frac{i}{2m\hbar} \int dt dt' \, J(t) \, G(t,t') \, J(t')} \right|_{J=0} = \frac{i\hbar^2}{2m\hbar} \left(G(t_2, t_1) + G(t_1, t_2) \right)$$

Found previously that with our $\omega^2 \to \omega^2 + i\varepsilon$ prescription

$$G(\tau = t_1 - t_2) = \theta(\tau)\,\frac{e^{-i\Omega\tau}}{2i\Omega} + \theta(-\tau)\,\frac{e^{i\Omega\tau}}{2i\Omega} = \frac{e^{-i\Omega|\tau|}}{2i\Omega}$$

$$\Rightarrow \langle 0|\hat{T}\{x(t_1)\,x(t_2)\}|0\rangle = \frac{i\hbar}{m}\,G(t_1,t_2) =$$

$$= \frac{\hbar}{2m\Omega}\,e^{-i\Omega|t_1-t_2|}$$

time-ordered Green's function

(Do this problem in operator formalism as an exercise)

* Same is true for ~~[illegible]~~ higher order GFs.

From this can find, for example, equal time matrix element:

$$\lim_{t_1\to t_2} \langle 0|\hat{T}\{\hat{x}(t_1)\,\hat{x}(t_2)\}|0\rangle = \langle 0|\hat{x}^2(t)|0\rangle = \frac{\hbar}{2m\Omega}.$$

Same via Fourier transform

$$Z(J) = N\int Dx \exp \frac{i}{\hbar}\int_{-\infty}^{\infty} dt \left(\frac{1}{2}m\dot{x}^2 - \frac{1}{2}m\Omega^2 x^2 + Jx\right)$$

$$x(t) = \int \frac{d\omega}{2\pi} e^{-i\omega t} x(\omega)$$

But $$\int_{-\infty}^{\infty} dt \int \frac{d\omega_1}{2\pi}\frac{d\omega_2}{2\pi} \underbrace{e^{-i\omega_1 t - i\omega_2 t}}_{2\pi\delta(\omega_1+\omega_2)} \underbrace{(-i\omega_1)(-i\omega_2)}_{-\omega_1\omega_2 \to \omega_1^2} x(\omega_1)x(\omega_2)$$

$$= \int_{-\infty}^{\infty} \frac{d\omega_1}{2\pi} x(\omega_1)x(-\omega_1)$$

Observe $$\begin{cases} x(+\omega) = \int dt\, e^{i\omega t} x(t) \\ x(-\omega) = \int dt\, e^{-i\omega t} x(t) = (x(\omega))^* \end{cases}$$

Similarly $$\int_{-\infty}^{\infty} dt\, x^2(t) = \int \frac{d\omega}{2\pi} x(\omega)x(-\omega)$$

$$\int_{-\infty}^{\infty} dt\, J(t)x(t) = \int_{-\infty}^{\infty}\frac{d\omega}{2\pi} \underbrace{\int_{-\infty}^{\infty} dt\, e^{-i\omega t} J(t)}_{J(-\omega)} x(\omega) = \int \frac{d\omega}{2\pi} J(-\omega)x(\omega)$$

$$\Rightarrow Z(J) = N\int Dx(\omega) \exp \frac{i}{\hbar}\int_{-\infty}^{\infty}\frac{d\omega}{2\pi} x(-\omega)\left[\frac{1}{2}m\omega^2 - \frac{1}{2}m\Omega^2 + J(-\omega)\right]x(\omega)$$

$$= N\int Dx(\omega) \exp \frac{i}{\hbar}\int_{-\infty}^{\infty}\frac{d\omega}{2\pi}\Big\{x(-\omega)\,\frac{1}{2}m(\omega^2 - \Omega^2 + i\varepsilon)\,x(\omega)$$

convergence factor!

$$+ \frac{1}{2}J(-\omega)x(\omega) + \frac{1}{2}J(\omega)x(-\omega)\Big\}$$

Now work to "complete the square" in the exponential:

$$\left(\frac{im}{2\hbar}\right)\int_{-\infty}^{\infty}\frac{d\omega}{2\pi}\left\{\left(\omega^2-\Omega^2+i\varepsilon\right)\left(x(\omega)+\frac{J(\omega)}{m(\omega^2-\Omega^2+i\varepsilon)}\right)\cdot\right.$$

$$\cdot\left(x(-\omega)+\frac{J(-\omega)}{m(\omega^2-\Omega^2+i\varepsilon)}\right)-$$

$$\left.-\frac{J(-\omega)}{m}\frac{1}{\omega^2-\Omega^2+i\varepsilon}\frac{J(\omega)}{m}\right\}$$

So we can shift $x(\omega)\to\eta(\omega)-\dfrac{J(\omega)}{m(\omega^2-\Omega^2+i\varepsilon)}$

This does not change integration measure!

$$Z(J)=N\int D\eta(\omega)\;e^{\frac{im}{2\hbar}\int\frac{d\omega}{2\pi}\eta(\omega)(\omega^2-\Omega^2+i\varepsilon)\eta(-\omega)}$$

$$\cdot\exp\left\{-\frac{i}{2m\hbar}\int_{-\infty}^{\infty}\frac{d\omega}{2\pi}J(-\omega)\frac{1}{\omega^2-\Omega^2+i\varepsilon}J(\omega)\right\}$$

Fourier back: $J(-\omega)=\int dt\;e^{-i\omega t}J(t)$

$J(\omega)=\int dt'\;e^{i\omega t'}J(t')$

But $\displaystyle\int\frac{d\omega}{2\pi}\int dt\,dt'\;J(t)J(t')\frac{e^{-i\omega(t-t')}}{\omega^2-\Omega^2+i\varepsilon}=$

$$=\int_{-\infty}^{\infty}dt\,dt'\;G(t-t')J(t)J(t')$$

$\Rightarrow$ again $Z(J)=Z(0)\;e^{-\frac{i}{2m\hbar}\int_{-\infty}^{+\infty}dt\,dt'\;G(t-t')J(t)J(t')}$